

ACRME Region Selection — Interactive Mockup: Design Plan

Status: Draft — Awaiting Approval

Date: 2026-09-21

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Sources: `acrme_calculation_logic_comprehensive_walkthrough.md`, `acrme_requirements_baseline_v2_4.md`, `acrme_plain_english_walkthrough.md`

Executive Summary

This plan defines an **interactive functional prototype** of the ACRME Region Selection engine. It is not a design wireframe — it is a working simulation backed by mock data that faithfully reproduces:

- The Hard Constraint Gate (HC-1..HC-10 subset, arithmetic-critical ones)
- All three placement scoring formulas (PS_Prod, PS_NonProd, PS_DR)
- The sequential Prod → CVAL → DR selection logic
- The geography distribution models (3-region US, 2-region co-located EU/AU/APAC, Middle East cross-geo DR)
- The DR max-not-sum sizing model (A.6, DR-017)

The tool's primary purpose is to **validate and communicate the algorithm** — enabling the team to test edge cases, tune weights, and demonstrate the selection logic without needing a live Azure environment.

1. Requirements

1.1 Functional Requirements

ID	Requirement
FR-01	User selects a geography and a customer SKU + VM count
FR-02	User can optionally supply an exact Prod region (default path) or leave it to engine selection (exception path)
FR-03	Tool runs HC-3, HC-6, HC-7 gates on every candidate region — displays pass/fail per constraint per region
FR-04	Tool scores all HC-passing candidates with PS_Prod, PS_NonProd, PS_DR — shows each component value (α , β , γ , δ , ϵ) and weighted contribution
FR-05	Tool selects CVAL and DR regions via argmax after Prod is fixed
FR-06	Tool shows final placement as a Customer-SeedRecord (Prod / CVAL / DR)
FR-07	Tool displays readiness state: READY , QUOTA_DEFICIT , CAPACITY_UNAVAILABLE , POLICY_BLOCKED
FR-08	User can edit scoring weights (α , β , γ , δ , ϵ) in a policy panel; tool enforces sum = 1.0 (± 0.001) and re-scores live
FR-09	Tool highlights which factor most influenced the winning region (“ α headroom drove this choice”)
FR-10	Mock data is editable (region snapshot values) — changes re-trigger the full scoring run
FR-11	Tool supports all five geographies: US (3-region), EU, AU, APAC, ME (with cross-geo DR)
FR-12	Tool surface the total_customers design gap with a selectable denominator mode

1.2 Non-Functional Requirements

ID	Requirement
NFR-01	All formulas implemented exactly as specified in baseline — no approximations
NFR-02	Components clamped to [0, 1] before weighting
NFR-03	Displayed arithmetic should be reproducible from the visible input values — no hidden state
NFR-04	The tool is a simulation only — no calls to Azure APIs
NFR-05	Weights editable via UI panel — config-driven as the baseline specifies

2. Mock Data Model

2.1 Geography Registry

The five in-scope geographies (REG-001, PLC-010a, PLC-010b):

Geography	Distribution Model	Standard Regions	DR Exception
US	3-region	West US 3 · Central US · Canada Central	None
EU	2-region co-located	Switzerland North · Sweden Central	None
Australia	2-region co-located	Australia East · Australia Southeast	None
Asia Pacific	2-region co-located	East Asia · Southeast Asia	None
Middle East	2-region cross-geo DR	UAE North · Saudi Arabia Central	DR → EU (weighted)

2.2 Regional Snapshot Schema

Each region carries the following snapshot fields (the engine's runtime inputs — all editable in the mockup):

```

interface RegionSnapshot {
  region_id: string;           // e.g. "westus3"
  region_name: string;         // e.g. "West US 3"
  geography: Geography;
  az_count: number;           // 2 or 3 (ε component)

  // Prod CRG
  prod_crg: {
    quantity: number;         // total reserved vCPU (α denominator for
  PS_Prod)
    allocated: number;         // vCPU in use by running VMs
    free_slots: number;        // quantity - allocated
    quota_limit: number;       // subscription vCPU cap (β denominator)
    quota_headroom: number;    // quota_limit - current_usage (β numerator)
  };

  // NonProd CRG
  nonprod_crg: {
    quantity: number;
    allocated: number;
    effective_free: number;    // free after DR overflow reserve is deducted
    quota_used: number;
    quota_limit: number;
    quota_headroom: number;
  };

  // DR CRG (max-not-sum sized)
  dr_crg: {
    quantity: number;         // max-not-sum of source workloads' vCPU
    free_slots: number;        // usable standby capacity
    coverage_ratio: number;    // 6 component (clamped [0,1])
    bootstrap_qty: number;     // HC-6 floor: min dr_crg.free + non-
  prod_eff_free
    dr_coverage_target: number; // scoring-only normalization (not a sizing
  ratio)
  };

  // Customer distribution
  prod_customer_count: number; // γ numerator
  // total_customers is geography-scoped (see Section 2.5)

  // Minimum floors (HC-3 absolute minimums)
  min_prod_headroom_vcpu: number; // default 20
  min_nonprod_headroom_vcpu: number; // default 20
  min_dr_headroom_vcpu: number; // default 16
}

```

2.3 Mock Region Data — US Geography

Pre-loaded starter values (derived from the comprehensive walkthrough document):

Field	West US 3	Central US	Canada Central
az_count	3	3	2
prod_crg.quantity	200	200	200
prod_crg.allocated	120	160	80
prod_crg.free_slots	80	40	120
prod_crg.quota_limit	1000	1000	1000
prod_crg.quota_headroom	700	550	300
non-prod_crg.quantity	120	100	80
non-prod_crg.allocated	80	100	60
non-prod_crg.effective_free	130	80	40
non-prod_crg.quota_headroom	240	160	328
non-prod_crg.quota_used	80	100	60
dr_crg.quantity	120	80	60
dr_crg.free_slots	100	60	60
dr_crg.coverage_ratio	0.50	0.45	0.60
dr_crg.bootstrap_qty	60	60	40
dr_crg.dr_coverage_target	0.80	0.80	0.80
prod_customer_count	40	25	20

2.4 Mock Region Data — EU Geography

Field	Switzerland North	Sweden Central
az_count	3	3
prod_crg.quantity	150	150
prod_crg.allocated	100	80
prod_crg.quota_limit	600	600
prod_crg.quota_headroom	400	420
nonprod_crg.effective_free	80	90
nonprod_crg.quota_headroom	100	138
dr_crg.quantity	80	80
dr_crg.free_slots	60	65
dr_crg.coverage_ratio	0.60	0.65
dr_crg.bootstrap_qty	40	40
prod_customer_count	6	4

2.5 Mock Region Data — Middle East + EU DR Destination

Middle East regions carry their own snapshot (Prod/CVAL only — DR goes cross-geo):

Field	UAE North	Saudi Arabia Central
az_count	3	2
prod_crg.quantity	100	100
prod_crg.allocated	40	30
prod_crg.quota_headroom	200	180
prod_customer_count	3	2

For Middle East DR: the tool runs PS_DR scoring over the **EU Standard regions** (Switzerland North · Sweden Central) as cross-geo DR candidates (PLC-010b, DR-020).

2.6 Australia and Asia Pacific — Default Starter Values

Both follow the same 2-region co-location pattern as EU. Pre-loaded with symmetric starter values (quantity=150, quota_limit=500, moderate utilization ~50%, 2-4 customers). These are illustrative and fully editable.

2.7 SKU Registry

SKU	vCPU per VM	Pre-loaded
E16ads_v5	16	✓
E8ads_v5	8	✓
D8s_v5	8	✓
D16s_v5	16	✓

2.8 The `total_customers` Design Gap — UI Treatment

The `total_customers` denominator in the γ component is **undefined in the baseline** (documented open issue in all three source documents). The mockup exposes this ambiguity explicitly and gives the user a chooser:

Mode	Denominator	Behaviour
Live geography total (default)	Sum of <code>prod_customer_count</code> across all Standard regions in the geography	Changes with every new placement; derived from snapshot — no external input
Policy constant	Configurable integer (e.g. 20 for US, 15 for EU)	Static target; represents maximum intended customers per geography
Manual entry	User types a value	Useful for reproducing the walkthrough examples exactly

A yellow annotation in the UI labels γ calculations with a footnote: “ $\triangle$ `total_customers` is an unresolved design term — see baseline open issue.”

3. Formulas to Implement

3.1 `PS_Prod(r)` — Production Placement Score

```

 $\alpha_{\text{component}}$  = Clamp(nonprod_crg.effective_free / prod_crg.quantity)
 $\beta_{\text{component}}$  = Clamp(prod_crg.quota_headroom / prod_crg.quota_limit)
 $\gamma_{\text{component}}$  = Clamp(1 - prod_customer_count / total_customers)
 $\delta_{\text{component}}$  = Clamp(dr_crg.coverage_ratio)
 $\epsilon_{\text{component}}$  = Clamp(az_count / 3)

 $\text{PS\_Prod}(r) = \alpha \times \alpha_{\text{component}} + \beta \times \beta_{\text{component}} + \gamma \times \gamma_{\text{component}} + \delta \times \delta_{\text{component}} + \epsilon \times \epsilon_{\text{component}}$ 

```

3.2 PS_NonProd(r) — CVAL/NonProd Placement Score

```

α_component = Clamp(nonprod_crg.effective_free / nonprod_crg.quantity)
β_component = Clamp(nonprod_crg.quota_headroom / nonprod_crg.quota_limit) [reuse Non-Prod quota pool]
γ_component = Clamp(1 - prod_customer_count / total_customers) [same γ - known design gap]
δ_component = Clamp(nonprod_crg.effective_free / nonprod_crg.quantity) [duplicate of α - known design gap]
ε_component = Clamp(az_count / 3)

PS_NonProd(r) = α×α_component + β×β_component + γ×γ_component + δ×δ_component + ε×ε_component

```

Design gap (surfaced in UI): For PS_NonProd, the δ component duplicates α (both use `nonprod_crg.effective_free / quantity`). The plain-english walkthrough flags this as “duplicate of α in design-of-record.” The mockup displays a note alongside the δ row.

3.3 PS_DR(r) — DR Placement Score

```

α_component = Clamp(dr_crg.free_slots / dr_crg.quantity)
β_component = Clamp(dr_crg.quota_headroom / dr_quota_limit)
γ_component = Clamp(1 - prod_customer_count / total_customers)
δ_component = Clamp(dr_crg.coverage_ratio / dr_crg.dr_coverage_target)
ε_component = Clamp(az_count / 3)

PS_DR(r) = α×α_component + β×β_component + γ×γ_component + δ×δ_component + ε×ε_component

```

3.4 Clamp Function

```
Clamp(x) = max(0, min(1, x))
```

All components are clamped before weighting. The mockup shows the raw ratio AND the clamped value when they differ.

3.5 Hard Constraint Gate

The tool checks these three arithmetic HCs before scoring. Regions that fail any are excluded:

HC-3 (Quota Floor):

```

For Prod:   prod_crg.quota_headroom ≥ (vm_count × vcpu_per_vm)
            AND prod_crg.quota_headroom - requested_vcpu ≥ min_prod_headroom_vcpu

For NonProd: nonprod_crg.effective_free ≥ (vm_count × vcpu_per_vm)
            AND nonprod_crg.quota_headroom - requested_vcpu ≥ min_nonprod_headroom_vcpu

For DR:     dr_crg.free_slots ≥ dr_bootstrap_qty (DR-007)

```

HC-6 (DR Coverage Floor):

```
dr_crg.free_slots + nonprod_crg.effective_free ≥ dr_crg.bootstrap_qty
```

HC-7 (DR Floor Integrity — NonProd gate):

```
nonprod_crg.quota_used + requested_vcpu ≤ nonprod_quota_ceiling - dr_crg.bootstrap_qty
```

HC-1, HC-2, HC-4, HC-5, HC-8, HC-9, HC-10, HC-11 are non-arithmetic (governance, policy, restricted-region checks). They are represented in the mockup as **toggle switches** on each region (“Restricted region?”, “Exception approval on file?”, “CRG sealed?”) — defaulting to values that represent the happy path.

3.6 DR Requirement: Max-Not-Sum (A.6, DR-017)

For any DR destination region, the DR capacity requirement is:

```
Destination_DR_Requirement(dr_region) = MAX over all source regions routing to
dr_region of:
    (source_region_customer_count × vcpu_per_deployment)
```

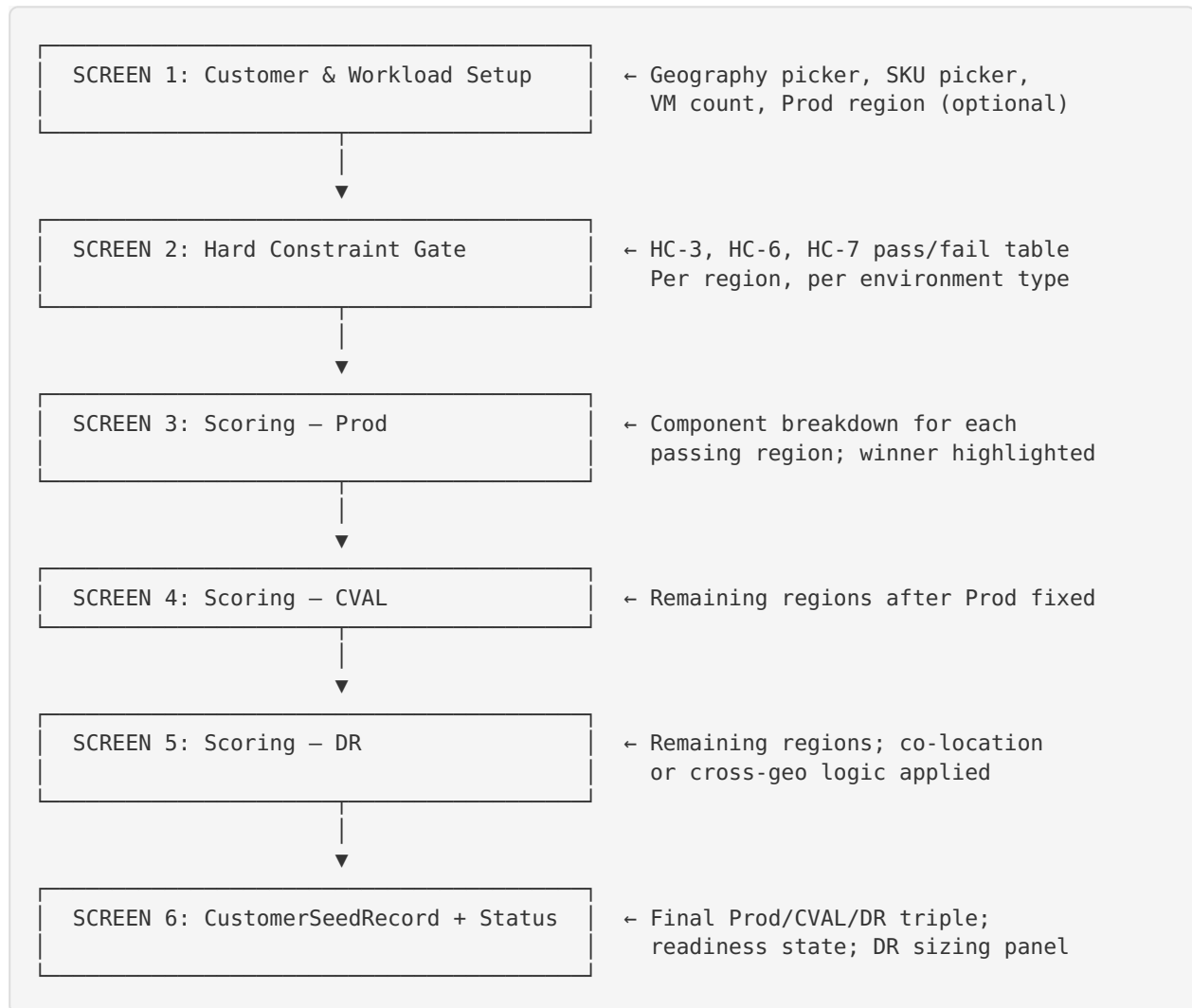
This value is shown in the DR sizing panel alongside the `dr_crg.quantity` and the gap/surplus.

3.7 Weight Invariant Validation

```
if |α + β + γ + δ + ε - 1.0| > 0.001:
    → PolicyValidationError: "Weights must sum to 1.0 ± 0.001"
    → Scoring is blocked until corrected
```

4. UI Design

4.1 Screen Flow



There is also a persistent **Policy Panel** (collapsible sidebar) available from any screen that exposes scoring weights (with live validation), `total_customers` mode, and the auto-increase threshold reference values.

4.2 Screen 1 — Customer & Workload Setup

Inputs:

- Geography (dropdown): US · EU · Australia · Asia Pacific · Middle East
- SKU (dropdown): E16ads_v5 · E8ads_v5 · D8s_v5 · D16s_v5
- VM count (stepper, default 3): vCPU = vm_count × vcpu_per_vm is computed and shown inline
- Prod region (optional text/dropdown):
- If supplied → default path (validate only, audit-score only)
- If blank → exception path (governance toggle must be enabled)
- Customer ID (free text, for seed record display)

Output: Workload summary card shown at top of all subsequent screens.

4.3 Screen 2 — Hard Constraint Gate

A table with one row per candidate region and columns for each HC check:

Region	HC-3 Prod	HC-6 DR Floor	HC-7 DR Integrity	Restricted?	Exception?	ELIGIBLE?
West US 3	✓ $700 \geq 48$	✓ $160 \geq 60$	✓ pass	✗ No	—	✓ ELIGIBLE
Central US	✓ $550 \geq 48$	✓ $140 \geq 60$	✓ pass	✗ No	—	✓ ELIGIBLE
Canada Central	✗ $300 < 48^*$	— (skipped)	— (skipped)	✗ No	—	✗ EXCLUDED

*HC-3 fails → remaining HCs are skipped (short-circuit evaluation).

Clicking any cell opens a drawer showing the full arithmetic:

```

HC-3 Prod: Canada Central
Requested:      48 vCPU (3 × E16ads_v5)
Quota headroom: 300 vCPU
Min floor:      20 vCPU
Check:          300 ≥ 48 → PASS
Post-allocation: 300 - 48 = 252 ≥ 20 → PASS
  
```

(This example uses the baseline walkthrough data, where Canada Central actually passes HC-3. The mock data above is set up to show a mix of passes and failures.)

For each excluded region, the UI shows the readiness state that would be returned: `QUOTA_DEFICIT`, `CAPACITY_UNAVAILABLE`, etc.

4.4 Screen 3 — Prod Scoring Table

For all HC-passing regions, the scoring table shows components side by side:

	West US 3	Central US
α (headroom)	$130/200 \rightarrow \mathbf{0.65} \times 0.30 = \mathbf{0.195}$	$80/200 \rightarrow \mathbf{0.40} \times 0.30 = \mathbf{0.120}$
β (quota)	$700/1000 \rightarrow \mathbf{0.70} \times 0.20 = \mathbf{0.140}$	$550/1000 \rightarrow \mathbf{0.55} \times 0.20 = \mathbf{0.110}$
γ (fairness)	$1-40/100 \rightarrow \mathbf{0.60} \times 0.25 = \mathbf{0.150}$	$1-25/100 \rightarrow \mathbf{0.75} \times 0.25 = \mathbf{0.188}$
δ (DR ready)	$\mathbf{0.50} \times 0.15 = \mathbf{0.075}$	$\mathbf{0.45} \times 0.15 = \mathbf{0.068}$
ϵ (zones)	$3/3 \rightarrow \mathbf{1.00} \times 0.10 = \mathbf{0.100}$	$3/3 \rightarrow \mathbf{1.00} \times 0.10 = \mathbf{0.100}$
PS_Prod	0.660 ← WINNER	0.586

A bar chart shows all five weighted contributions stacked for each region.

A “Margin of victory” line shows the score gap and which component most influenced the outcome.

If the user supplied a Prod region (default path), that region is shown in gold with the label “Customer-supplied — validated only” and the score is computed for audit.

4.5 Screen 4 & 5 — CVAL and DR Scoring

Same layout as Screen 3, but:

- CVAL scoring excludes the Prod winner from the candidate pool
- DR scoring applies PLC-010a co-location logic (2-region geographies: DR must land in the same region as CVAL) or PLC-010b cross-geo (Middle East: DR candidate pool = EU Standard regions)
- DR scoring uses PS_DR formula, showing the δ component as $\text{coverage_ratio} / \text{dr_coverage_target}$

For a 2-region geography (EU/AU/APAC) where CVAL is deterministic (only one region left), the scoring table is replaced by a notice: “CVAL: deterministic — Sweden Central is the only remaining Standard region.”

4.6 Screen 6 — CustomerSeedRecord + Readiness

CustomerSeedRecord	
Customer: CUST-0042	
Geography: US	
● Prod:	West US 3 (48 vCPU, score 0.66)
● CVAL:	Canada Central (48 vCPU, score 0.69)
● DR:	Central US (48 vCPU, score 0.60)
Readiness State: READY	
Policy Version: v2.4.0-alpha	

DR Sizing Panel (below the seed record):

DR Capacity Sizing (max-not-sum, A.6):

Source: Central US → DR: West US 3
 Customers in Central US: $25 \times 48 \text{ vCPU} = 1200 \text{ vCPU}$ at risk

Source: Canada Central → DR: West US 3
 Customers in Canada Central: $20 \times 48 \text{ vCPU} = 960 \text{ vCPU}$ at risk

DR_Requirement(West US 3) = $\text{MAX}(1200, 960) = 1200 \text{ vCPU}$

DR_CRG quantity: 120 vCPU (starter – scaled for illustration)

Gap: $1200 - 120 = 1080 \text{ vCPU}$ Δ (auto-increase would trigger)

Overcommit ratio: $(1200 + 960) / 1200 = 1.80$

Note: the DR sizing numbers above use illustrative scale. The mockup will calculate from the actual mock data — showing the formula and its inputs.

5. Demo Scenarios

The tool pre-loads five scenarios accessible via a “Scenarios” button. Each sets the mock data state so the chosen path is immediately visible:

Scenario A — US Happy Path (3-region, all separate)

- Geography: US, SKU: E16ads_v5 × 3 VMs
- All three US Standard regions pass all HCs
- Result: Prod/CVAL/DR in three separate regions (full separation)
- Shows: scoring margin, the α component driving the winner

Scenario B — US Quota Deficit (HC-3 fail)

- Central US quota_headroom artificially set to 12 vCPU
- Central US fails HC-3; excluded from Prod scoring pool
- Two-candidate scoring race; readiness state = `QUOTA_DEFICIT` for that region

Scenario C — EU 2-Region Co-location (PLC-010a)

- Geography: EU, SKU: E8ads_v5 × 3 VMs
- Shows: Prod scoring, deterministic CVAL, mandatory DR co-location with CVAL
- Highlights co-location risk note (both CVAL and DR lost if Sweden Central fails)

Scenario D — EU Capacity Unavailable (both regions too small)

- Switzerland North headroom = 30 vCPU, E16ads_v5 needs 48 vCPU → HC-3 fail
- Sweden Central takes Prod; but CVAL check on Switzerland North also fails
- Readiness: `CAPACITY_UNAVAILABLE`

Scenario E — Middle East Cross-Geo DR (PLC-010b)

- Geography: Middle East, SKU: E8ads_v5 × 3 VMs
- Prod/CVAL co-locate in UAE North or Saudi Arabia Central (weighted selection)
- DR scores EU Standard regions via PS_DR
- Shows: how the cross-geo DR candidate pool differs from the ME pool

Scenario F — Weight Tuning (Distribution-First)

- Same data as Scenario A (US Happy Path)
- User opens Policy Panel and shifts weights to $\gamma=0.35$, $\alpha=0.25$ (distribution-first Option 1)
- Tool re-scores live; winner may change
- Demonstrates how weight policy changes affect placement

6. Implementation Approach

6.1 Architecture Decision

Recommended: Next.js functional prototype with JSON mock data

Rationale:

- Platform-hosted preview available (Abacus.AI infrastructure)
- No Azure API calls needed — all data is in-memory from JSON mock files
- React state management handles live re-scoring when weights or data change
- The engine logic (HC gate + scoring formulas) runs client-side in TypeScript — fast, reproducible
- The mockup can be shared with stakeholders as a URL, not a local file

Not recommended:

- Pure HTML/JS single-file: fine for simple demos, but the five-screen flow and live re-scoring need

proper state

- Python backend: adds latency for what is pure computation; unnecessary for a prototype

6.2 Project Structure

```

/acrme-region-selection-mockup
  /data
    regions.json      ← mock RegionSnapshot for all 5 geographies
    skus.json         ← SKU registry
    scenarios.json    ← pre-loaded scenario configurations
    policy.json       ← default PlacementPolicy (weights, floors, etc.)
  /lib
    engine/
      hardConstraints.ts ← HC-3, HC-6, HC-7 implementations
      scoring.ts        ← PS_Prod, PS_NonProd, PS_DR + Clamp
      selectionLogic.ts ← Prod→CVAL→DR sequential selection + geography rules
      drSizing.ts       ← max-not-sum DR requirement calculation
      policyValidator.ts ← weight-sum invariant check ( $\pm 0.001$ )
      types.ts         ← RegionSnapshot, PlacementPolicy, SeedRecord interfaces
    /components
      WorkloadSetup.tsx ← Screen 1
      HCGateTable.tsx   ← Screen 2
      ScoringTable.tsx  ← Screens 3, 4, 5
      SeedRecord.tsx    ← Screen 6 + DR sizing panel
      PolicyPanel.tsx   ← Collapsible sidebar: weight editor, total_customers mode
      ComponentBreakdown.tsx ← Stacked bar chart per component
  /app
    page.tsx          ← Main orchestration
  
```

6.3 Data Flow

```

User Input (Screen 1)
  ↓
policyValidator.ts  — validates weight sum
  ↓
hardConstraints.ts  — filters candidate pool per HC check (returns pass/fail +
reason per region)
  ↓
scoring.ts         — computes PS_Prod/NonProd/DR for passing candidates
  ↓
selectionLogic.ts  — Prod = argmax(PS_Prod); CVAL = argmax(PS_NonProd) over re-
remaining; DR = argmax(PS_DR) over remaining (or co-location rule for 2-region)
  ↓
drSizing.ts        — max-not-sum DR_Requirement for the selected DR region
  ↓
CustomerSeedRecord — {production_region, cval_region, dr_region, readiness_state,
policy_version}
  
```

All computation is synchronous and runs on state change — no async calls.

6.4 Mock Data Editability

The **Policy Panel** sidebar allows:

1. **Weight editor:** α , β , γ , δ , ϵ sliders (0.00–1.00, step 0.01) with a live sum display. Weight sum validation runs on every change; error banner shown if out of tolerance.
2. **total_customers mode:** radio buttons (Live Geography Total / Policy Constant / Manual Entry)
3. **Region data editor:** expandable cards for each region with editable snapshot fields. All arithmetic in Screens 2–6 re-runs on field change.
4. **Scenario loader:** dropdown that resets all mock data to a named scenario state.

6.5 Design Gaps Surfaced in UI

The following known gaps from the source documents are **annotated inline**, not hidden:

Gap	Location in UI	Display Treatment
<code>total_customers</code> undefined	γ component row in scoring table	Yellow  icon with tooltip linking to the open issue description
PS_NonProd δ duplicates α	δ row in NonProd scoring table	Blue  icon: “Known design gap — δ mirrors α for Non-Prod. Baseline v2.4 does not yet resolve this.”
<code>dr_bootstrap_qty</code> is TBD	HC-6 gate row	Field shows default (configurable), annotation: “Baseline value is TBD per workload”
<code>dr_coverage_target</code> is TBD	δ row in PS_DR table	Same annotation pattern

7. Data Dimensions

Dimension	Count	Rationale
Geographies	5	All in-scope (US, EU, AU, APAC, ME)
Total Standard regions	11	US(3) + EU(2) + AU(2) + APAC(2) + ME(2)
Mock customers per region	2-40	Enough to make γ non-trivial
SKUs	4	E16ads_v5, E8ads_v5, D8s_v5, D16s_v5
Pre-loaded scenarios	6	A-F (see Section 5)
VM count options	1-10 (stepper)	$\text{vCPU} = \text{vm_count} \times \text{vcpu_per_vm}$

This is **sufficient to exercise all formula paths** without overwhelming the UI. If the team wants to extend to more regions or customer volumes later, the JSON data file can be expanded without code changes.

8. Open Questions (Require Resolution Before or During Build)

#	Question	Impact	Recommendation
OQ-1	<code>total_customers</code> definition — which interpretation does the team want modelled as the default in the mockup?	γ values change materially	Default to “Live Geography Total” (sum of <code>region_customer_counts</code>); expose all modes via the chooser
OQ-2	PS_NonProd δ component — is the duplicate of α intentional or a spec gap to be resolved?	Affects NonProd scoring fidelity	Show duplicate as-is with annotation; flag as design gap for discussion
OQ-3	<code>dr_bootstrap_qty</code> concrete values — what should the mock data use?	HC-6 pass/fail depends on this	Use 40–60 vCPU range in mock data (lean, per baseline guidance ~5–20%); make configurable
OQ-4	<code>dr_coverage_target</code> concrete values — what normalization target for δ in PS_DR?	$\delta_{\text{component}} = \text{coverage_ratio} / \text{target}$	Use 0.80 as starter in mock data (editable)
OQ-5	Middle East DR destination set — exact EU Standard regions the engine selects from?	DR scoring candidate pool	Use {Switzerland North, Sweden Central} as the mock EU destination set
OQ-6	Should the mockup include the quota pool arithmetic (Part 2 of walkthroughs) as a separate panel?	Scope increase	Out of scope for v1 — Region Selection only; quota management is a follow-on mockup

9. Delivery Plan

Phase 1 — Region Selection Prototype (this build)

- All 5 geographies loaded with mock data
- HC gate (HC-3, HC-6, HC-7)

- PS_Prod / PS_NonProd / PS_DR scoring
- Prod → CVAL → DR sequential selection
- CustomerSeedRecord + readiness state output
- Policy Panel (weights, total_customers mode)
- 6 pre-loaded demo scenarios
- Design gaps annotated inline

Phase 2 — Quota Management Panel (follow-on)

- Pool_Limit calculation
- Allocatable_NonProd
- HC-7 pool arithmetic
- QUOTA_DEFICIT / CAPACITY_UNAVAILABLE simulation

Phase 3 — Capacity Reconciliation Simulator (follow-on)

- Target vs reserved comparison
 - Auto-increase threshold triggers
 - Max-not-sum DR requirement calculator (extended)
 - Reconciliation loop visualization
-

10. Decision Log

Decision	Alternative Considered	Rationale	Trade-off
Next.js functional prototype (not wireframes)	Figma/design mockup	User requested a “mock selection tool based on mocked up data” — implies interactive computation, not static screens	Slightly more build time; produces a demo that actually runs the formulas
All computation client-side	Python/Node API backend	No async latency; mock data fits in memory; no Azure calls needed	Cannot simulate real Azure API delays (acceptable for a prototype)
JSON mock data files	Hardcoded constants in components	Makes mock data editable from the UI without code changes	Requires a JSON schema; small overhead
6 pre-loaded scenarios	User-defined scenarios only	Ensures core paths are always demonstrable; reduces setup friction	Scenarios may not cover every edge case the team invents
Design gap annotations inline	Silent gaps (pretend they don’t exist)	The team needs to see where the spec is incomplete during prototyping — hiding gaps delays resolution	Slightly more complex UI
Scope: Region Selection only (Phase 1)	Include Quota + Capacity in one build	Quota and Capacity add substantial complexity; Region Selection can be validated independently first	Follow-on builds required for full ACRME coverage

End of Plan — Awaiting approval before implementation begins.